150 Hackathon Statements

II Year B.Tech CSE (Data Science)

1. Crop Yield Prediction System Design a system that predicts crop yield for a given region using historical rainfall, temperature, soil pH, and fertilizer usage data. The system should apply regression models (linear/polynomial) from the data science pipeline to estimate output per acre, while an underlying Java OOP module manages farmer profiles, land records, and crop history using class hierarchies. The tool should help agricultural officers plan procurement and storage in advance, reducing post-harvest wastage. Visualize predicted vs actual yield trends over multiple seasons to build farmer trust in the system.

2. Smart Irrigation Scheduler Build a water distribution optimizer for a village's farmland using graph algorithms (Prim's/Kruskal's MST or shortest-path) to minimize pipeline loss and ensure equitable water sharing among multiple plots. Incorporate soil moisture sensor data and evapotranspiration rates to dynamically adjust irrigation schedules using time-series analysis. The OOP design should represent each farm plot as an object with attributes like crop type, area, and water requirement. The end goal is reducing groundwater over-extraction in drought-prone mandals of Andhra Pradesh.

3. Farmer-to-Market Price Discovery Platform Create a real-time mandi price lookup and comparison system using hash tables and balanced BSTs for fast search across hundreds of markets and commodities. Scrape/aggregate public agri-market datasets and apply trend analysis to alert farmers when prices are favorable for selling. Implement an OOP-based notification and subscription system so farmers can track specific crops. This directly tackles the problem of middlemen exploitation and information asymmetry in rural markets.

4. Soil Health Classification System Develop a soil health card generator that classifies soil samples into fertility categories using clustering (K-means) on NPK, pH, and micronutrient datasets from government soil testing labs. The system should recommend suitable crops and fertilizer dosages based on the cluster the sample falls into. Use OOP to model different soil types as class hierarchies with inherited fertilization rules. The output should be a simple, vernacular-friendly report farmers can act on immediately.

5. Pesticide Usage Optimizer Build a decision-support tool that minimizes pesticide cost while maximizing crop protection using dynamic programming to solve a constrained optimization problem across multiple pest threats and budget limits. Integrate historical pest-outbreak data with weather patterns to predict pest risk windows using classification models. Design pesticide and crop objects using OOP inheritance to handle different chemical compatibility rules. This reduces chemical overuse, protecting both soil health and farmer expenditure.

6. Crop Rotation Planner Create a multi-season crop rotation recommender that uses greedy algorithms to sequence crops based on soil nutrient depletion and replenishment cycles, informed by data science models trained on historical soil nutrient datasets. The system must track nitrogen-fixing vs nitrogen-depleting crops and suggest optimal rotation cycles to sustain long-term soil fertility. OOP class design should represent crops with nutrient-impact attributes for extensibility. Field officers can use this to counsel farmers against mono-cropping-induced soil degradation.

7. Rural Supply Chain Route Optimizer Design a farm-to-warehouse logistics optimizer using Dijkstra's or A* algorithm to find the shortest, least-cost delivery routes for perishable produce across rural road networks. Incorporate real or simulated traffic and road-quality data to adjust route weights dynamically, and use regression to predict spoilage risk based on transit time and temperature. Model trucks, warehouses, and farms as OOP entities with capacity constraints. The objective is reducing the 20-30% post-harvest loss common in Indian agri-supply chains.

8. Livestock Disease Outbreak Predictor Build a predictive system for livestock disease outbreaks using time-series analysis of veterinary hospital records, vaccination logs, and seasonal patterns. Apply classification algorithms to flag high-risk zones and livestock categories before an outbreak spreads. Use OOP to model different animal types and their disease susceptibility profiles for extensibility to new diseases. This tool can help veterinary departments proactively allocate vaccines and medical camps in high-risk rural clusters.

9. Seed Quality Classification System Develop an automated seed quality grading tool using decision tree classifiers trained on physical attributes such as size, color, moisture content, and germination rate from agricultural lab datasets. The system should categorize seed batches into quality grades to prevent farmers from purchasing substandard or counterfeit seeds. Use OOP to represent seed lots, suppliers, and certification authorities as interacting classes. This addresses the widespread issue of spurious seed sales in rural markets.

10. Groundwater Level Depletion Tracker Create a groundwater monitoring dashboard that uses regression models to predict depletion trends across mandals based on rainfall, extraction rate, and cropping pattern datasets. Implement a trie-based search structure for fast village/mandal name lookups across thousands of monitoring stations. The OOP layer should manage borewell records, ownership, and historical readings. This tool supports district administrations in enforcing groundwater regulation in critically over-exploited blocks.

11. Agricultural Loan Eligibility & Risk Scoring System Build a rule-based loan eligibility engine for farmers using OOP design patterns (strategy/rule-chain) to evaluate multiple criteria like land holding, crop type, past repayment history, and weather risk. Apply classification models to assign a risk score predicting default probability, trained on anonymized cooperative bank datasets. The system should generate a transparent, explainable report for loan officers. This aims to improve credit access for genuine farmers while reducing NPAs for rural banks.

12. Weather-Based Crop Insurance Claim Automation Design an automated crop insurance claim verification system that uses threshold-based algorithms on rainfall, temperature, and humidity data to detect qualifying weather events (drought, excess rain) without manual field surveys. Apply anomaly detection to flag fraudulent claims that don't match regional weather patterns. Model policies, farmers, and claims using OOP inheritance for different insurance schemes. This can drastically cut down claim settlement time under schemes like PMFBY.

13. Farm Equipment Sharing/Rental Matching System Create a peer-to-peer farm equipment rental platform that uses bipartite matching algorithms to pair equipment owners with nearby farmers needing tractors, harvesters, or sprayers during peak seasons. Apply clustering to identify demand hotspots by season and geography for better equipment placement. Use OOP to model equipment, owners, and renters with booking and pricing logic. This reduces capital burden on small farmers who cannot afford individual machinery.

14. Post-Harvest Storage Loss Predictor Build a predictive model that estimates storage loss risk for different crops based on humidity, temperature, and pest presence data in warehouses, using classification algorithms trained on historical spoilage records. Alert warehouse managers via a priority-queue-based ranking system when intervention is urgently needed for specific storage units. Design storage units and crop batches as OOP objects with shelf-life attributes. This directly reduces the massive food wastage occurring in unregulated rural storage.

15. Multilingual Crop Advisory Chatbot Backend Develop the backend logic for a crop advisory chatbot that uses basic NLP techniques to parse farmer queries in regional languages and route them using a priority queue based on urgency (e.g., pest attack vs general query). Integrate a knowledge base built from agricultural extension datasets to generate rule-based responses. Use OOP to structure different advisory modules (irrigation, pest, fertilizer) as interchangeable strategy classes. This bridges the information gap for farmers without direct access to agricultural experts.

16. Hospital Bed Allocation Optimizer Design a real-time bed allocation system for government hospitals that uses priority queues to rank incoming patients by medical urgency and greedy scheduling to assign available beds across departments during surges like epidemics. Apply classification models trained on triage data to auto-suggest urgency levels from symptom inputs. Use OOP to represent wards, beds, and patients with state transitions (admitted, discharged, critical). This can significantly reduce chaos and delays during health emergencies in overcrowded public hospitals.

17. Disease Outbreak Spread Simulator Build a simulation tool that models disease spread through a community using graph traversal algorithms (BFS/DFS) over a contact network, where nodes represent individuals and edges represent interactions. Combine this with data science models estimating transmission probability from public epidemiological datasets to simulate outbreak curves under different intervention scenarios. Use OOP to define person, location, and interaction classes for extensibility. The simulator can help local health departments plan containment zones and resource deployment.

18. Blood Bank Inventory & Donor Matching System Create a blood bank management system using hash maps for O(1) blood-type lookups and OOP inheritance to model different donor categories (regular, emergency, first-time). Apply time-series forecasting to predict blood shortage periods based on historical donation and demand patterns. The system should auto-notify compatible donors during critical shortages using a matching algorithm. This addresses frequent blood shortages faced by rural and semi-urban blood banks.

19. Ambulance Dispatch Route Optimizer Develop an ambulance routing system using Dijkstra's algorithm with dynamically weighted edges representing live traffic and road conditions to minimize response time during emergencies. Integrate historical accident/call data to predict high-demand zones using clustering, enabling smarter ambulance pre-positioning. Model ambulances, hospitals, and incidents as OOP entities with status tracking. This tackles the critical golden-hour delay problem in emergency medical response, especially in tier-2/3 cities.

20. Patient Queue Management System for Rural Clinics Build a queue management system for primary health centers that uses priority scheduling algorithms to balance walk-in patients, appointments, and emergency cases fairly and efficiently. Apply basic predictive analytics on historical footfall data to forecast peak hours and recommend staffing adjustments. Use OOP to model doctors, patients, and appointment slots with clear state management. This reduces long, unmanaged waiting times that discourage rural patients from seeking timely care.

21. Diabetes Risk Prediction System Create a diabetes risk assessment tool using logistic regression trained on lifestyle, dietary, and basic clinical parameters from publicly available health survey datasets. The system should output a risk percentage along with explainable contributing factors to encourage preventive action. Use OOP to structure patient profiles and risk-factor categories for future extension to other chronic diseases. This supports community health workers conducting screening camps in underserved areas.

22. Medicine Expiry & Stock Management System Design a pharmacy inventory system that uses a min-heap to always surface medicines nearest to expiry for priority dispensing (FEFO — First-Expiry-First-Out), reducing wastage. Apply demand forecasting using time-series models to predict restocking needs for essential medicines. Represent medicines, batches, and suppliers using OOP class hierarchies. This is especially valuable for government dispensaries that frequently face both stockouts and expired medicine wastage simultaneously.

23. Mental Health Sentiment Tracker Build a sentiment analysis tool that processes anonymized text responses from mental health surveys or helpline chat logs using basic NLP techniques to identify trends in stress, anxiety, or depression indicators across a student or employee population. Aggregate results using statistical methods to generate periodic wellbeing reports without compromising individual privacy. Use OOP to modularize different sentiment-scoring strategies. This can help institutions proactively design mental health support programs based on real trend data.

24. Vaccine Distribution Optimizer Develop a cold-chain vaccine distribution planner using network flow algorithms to optimize the movement of temperature-sensitive vaccines from central storage to remote health centers with minimal wastage. Combine this with demand prediction models based on population and historical uptake data. Use OOP to represent storage facilities, transport vehicles, and vaccine batches with temperature-tracking attributes. This directly supports large-scale immunization drives in geographically challenging rural terrains.

25. Malnutrition Detection in Children Create a malnutrition screening tool using classification algorithms on anthropometric data (height, weight, age) benchmarked against WHO growth standards to flag at-risk children in Anganwadi records. Apply clustering to identify malnutrition hotspot villages for targeted nutrition intervention programs. Use OOP to model child health records with growth-tracking history. This supports ICDS workers in early identification and intervention, which is critical for reducing child mortality and stunting.

26. Hospital Readmission Predictor Build a predictive model using decision trees or random forests trained on de-identified electronic health record patterns to estimate a patient's risk of readmission within 30 days of discharge. Identify key contributing factors (comorbidities, follow-up compliance) to help hospitals design better discharge planning. Use OOP to model patient history and treatment episodes as linked objects. This can reduce healthcare costs and improve patient outcomes in resource-constrained public hospitals.

27. Doctor-Patient Appointment Scheduling System Design an appointment scheduling system using interval scheduling algorithms to maximize doctor utilization while minimizing patient wait times across multiple specializations and time slots. Apply basic forecasting to predict no-show probability based on historical patient behavior, allowing smart overbooking. Model doctors, patients, and slots using OOP with conflict-resolution logic. This improves efficiency in busy government hospital OPDs where manual scheduling causes significant bottlenecks.

28. Air Quality vs Respiratory Illness Correlation Analyzer Build a data analysis tool that correlates public AQI datasets with regional respiratory illness hospital admission data using regression analysis to quantify health impact of pollution. Visualize trends across seasons and localities to identify high-risk zones and time periods. Use OOP to structure data sources and analysis pipelines for reusability across cities. This evidence can support local pollution control policies and public health advisories.

29. Elderly Fall-Risk Prediction System Create a fall-risk prediction tool using time-series classification on wearable sensor data (simulated or public datasets) capturing gait, balance, and movement patterns of elderly individuals. Flag high-risk individuals for preventive intervention like physiotherapy or home modifications. Use OOP to model sensor readings, individuals, and risk assessments as connected entities. This supports elderly care facilities and family caregivers in preventing serious fall-related injuries.

30. Organ Donation Matching System Develop an organ donor-recipient matching system using graph-based compatibility algorithms that consider blood type, tissue matching, geographic proximity, and urgency to optimize allocation fairness and speed. Apply priority-based ranking similar to real transplant allocation policies. Use OOP to model donors, recipients, organs, and hospitals with clear compatibility rules. This addresses India's critical organ donation shortage and allocation transparency issues.

31. Epidemic Contact-Tracing Simulation Build a contact-tracing simulation using union-find (disjoint set) data structures to efficiently group individuals into connected clusters of potential exposure during an epidemic. Combine with basic probability models to estimate infection spread likelihood within each cluster. Use OOP to represent individuals, locations, and exposure events. This demonstrates how digital contact-tracing systems, like those used during COVID-19, can be built and scaled efficiently.

32. Telemedicine Load-Balancer Design a load-balancing system for a telemedicine platform that distributes incoming patient consultation requests across available doctors using OOP polymorphism to handle different doctor specialization types. Apply queuing theory and simple predictive models to estimate wait times and optimize doctor allocation during peak hours. This ensures equitable and efficient access to virtual healthcare, especially critical for patients in remote areas with no local specialists.

33. Nutritional Deficiency Mapper Create a regional nutritional deficiency mapping tool using clustering algorithms on dietary survey and anemia/deficiency prevalence datasets to identify geographic patterns. Visualize results on an interactive map to guide targeted fortification or supplementation programs. Use OOP to structure survey data collection and regional aggregation modules. This supports public health planners in efficiently allocating nutrition intervention budgets to the highest-need districts.

34. Medical Inventory Fraud Detection Build an anomaly detection system that flags suspicious patterns in medical inventory transactions (unusual stock movements, mismatched billing) in government hospitals using statistical outlier detection methods. Use OOP to model transactions, inventory items, and audit trails for traceability. This tool can help curb pilferage and fraud in public healthcare supply chains, ensuring medicines actually reach patients who need them.

35. Health Camp Resource Allocator Design a resource allocation tool for free health camps that uses knapsack-style optimization to allocate a limited budget across medicines, staff, and equipment based on expected patient demographics and disease prevalence in the target area. Use OOP to model resource categories with cost-benefit attributes. This helps NGOs and government bodies maximize health impact per rupee spent in underserved rural and tribal areas.

36. Traffic Signal Optimization System Build a traffic signal timing optimizer using graph algorithms to model intersections and apply real-time or simulated congestion data to dynamically adjust green-light durations, reducing average wait time across a city grid. Use regression to predict traffic volume by time-of-day and day-of-week for proactive signal planning. Model intersections, signals, and traffic flows using OOP. This addresses chronic urban congestion that costs cities significant productivity and fuel every day.

37. Public Transport Route Planner Create a multi-modal public transport route planner using Dijkstra's or Floyd-Warshall algorithm to compute the shortest and most cost-effective paths across bus and metro networks with multiple stops and transfers. Incorporate real fare and schedule datasets to give accurate journey estimates. Use OOP to model routes, stops, and vehicles with schedule-based state. This improves commuter experience in cities where transport information is fragmented and hard to navigate.

38. Waste Collection Route Optimizer Design a garbage truck route optimization system using Travelling Salesman Problem (TSP) heuristics (nearest neighbor, 2-opt) to minimize total distance traveled while covering all designated collection points in a ward. Incorporate bin-fill-level data to prioritize routes dynamically. Use OOP to model trucks, bins, and routes with capacity constraints. This reduces fuel costs and improves waste collection efficiency for municipal corporations struggling with irregular collection schedules.

39. Parking Slot Allocation System Build a smart parking allocation system using greedy and priority-based scheduling algorithms to assign available slots to incoming vehicles based on arrival time, duration, and vehicle type. Apply demand forecasting using historical occupancy data to predict peak parking congestion periods. Use OOP to model parking lots, slots, and vehicles with booking states. This reduces the time and fuel wasted circling for parking in congested urban commercial areas.

40. Smart Streetlight Energy Optimizer Create an energy optimization system for streetlights that uses clustering on simulated occupancy/motion sensor data to identify low-traffic zones and time periods where dimming is safe, reducing energy consumption. Apply time-series analysis to correlate lighting needs with seasonal daylight variation. Use OOP to model streetlight units with adjustable brightness states. This can significantly cut municipal electricity costs while maintaining public safety standards.

41. Urban Flood Risk Predictor Build a flood risk prediction model using regression analysis on rainfall intensity, drainage capacity, and elevation datasets to identify flood-prone urban zones before monsoon season. Visualize risk levels on a city map to guide preventive drainage maintenance and public advisories. Use OOP to structure zones, drainage infrastructure, and risk assessments. This is critical for cities facing recurring urban flooding due to poor drainage planning and unplanned construction.

42. Water Pipeline Leak Detection System Design a leak detection system that applies anomaly detection algorithms on simulated flow-rate and pressure sensor data across a water distribution network to identify likely leak locations without physical inspection. Use graph algorithms to trace the pipeline network and isolate affected segments efficiently. Model pipeline segments and sensors using OOP. This addresses the massive non-revenue water loss (30-40% in many Indian cities) caused by undetected leaks.

43. City-Wide Crime Hotspot Predictor Create a crime hotspot prediction tool using spatial clustering (DBSCAN) on historical, anonymized incident location and time datasets to identify high-risk zones and time windows. Visualize results as heatmaps to support smarter police patrol allocation. Use OOP to structure incident records, zones, and patrol units. This data-driven approach can improve resource deployment efficiency for urban police departments with limited patrol capacity.

44. Building Fire Evacuation Route Planner Build an evacuation route planning tool using BFS/DFS on a graph representation of a building's floor plan to compute the fastest evacuation paths from any room to the nearest exit, accounting for blocked routes during simulated fire scenarios. Use OOP to model rooms, exits, and hazard zones as graph nodes with dynamic states. This tool can be integrated into building safety systems for schools, hospitals, and public buildings.

45. Public Grievance Redressal Ticketing System Design a citizen grievance management system using a priority-queue-based ticketing engine that ranks complaints (water, sanitation, roads) by severity and duration pending, ensuring urgent issues are addressed first. Apply basic NLP classification to auto-categorize complaints from free-text descriptions. Use OOP to model tickets, departments, and citizens with status-tracking workflows. This improves transparency and accountability in municipal grievance redressal systems.

46. E-Toll Traffic Flow Analyzer Build a traffic flow analysis tool using time-series analysis on vehicle count data at toll plazas to identify congestion patterns, predict peak travel times, and recommend dynamic toll pricing to smooth demand. Use OOP to model toll booths, vehicle categories, and pricing rules. This helps highway authorities reduce bottlenecks at toll plazas, a major source of highway congestion in India.

47. Noise Pollution Mapping System Create a noise pollution mapping tool that aggregates simulated or public sensor data across a city into a spatial heatmap using statistical aggregation methods, identifying noise violation hotspots near hospitals and schools. Apply time-series analysis to identify peak noise periods by location. Use OOP to model sensors, zones, and permissible-limit rules. This supports urban planning bodies in enforcing noise pollution control regulations more effectively.

48. Housing Price Prediction Model Build a real estate price prediction system using multiple linear regression trained on features like location, area, amenities, and proximity to infrastructure from public real estate datasets. Provide transparent, explainable price estimates to help first-time buyers avoid overpaying due to information asymmetry. Use OOP to structure property listings and valuation logic. This promotes fairness and transparency in a market often driven by opaque, negotiation-based pricing.

49. Slum Redevelopment Resource Allocator Design a resource allocation tool for slum redevelopment projects using constraint-based optimization to balance limited government budgets across housing, sanitation, and infrastructure needs based on population density and need-severity data. Use OOP to model redevelopment zones and resource categories with priority weighting. This supports urban local bodies in making data-driven, equitable redevelopment decisions rather than politically influenced ones.

50. Smart Bin Fill-Level Monitoring & Dispatch System Build a waste bin monitoring system using a min-heap to prioritize collection trucks toward the fullest bins first, based on simulated fill-level sensor data across a city. Combine with route optimization to minimize travel distance for collection rounds. Use OOP to model bins, trucks, and collection zones. This prevents overflowing bins and reduces unnecessary trips to partially-full bins, improving sanitation and operational efficiency.

51. Metro/Bus Fare Calculator with Dynamic Pricing Create a fare calculation engine using the OOP strategy design pattern to support multiple pricing models (distance-based, zone-based, peak/off-peak) for public transport systems. Apply demand data analysis to recommend dynamic peak-hour pricing that balances revenue and ridership. Use graph algorithms to compute distance-based fares along transit routes. This helps transport authorities design fairer, demand-responsive fare structures.

52. Encroachment Detection Using GIS Data Classification Build a basic land encroachment detection tool using classification techniques on simulated or public satellite/GIS boundary datasets to flag mismatches between registered land boundaries and actual land use. Use OOP to model land parcels, ownership records, and encroachment flags. This supports municipal survey departments in identifying unauthorized construction and land disputes more systematically.

53. City Budget Allocation Simulator Design a municipal budget allocation simulator using dynamic programming (a knapsack variant) to optimally distribute a fixed annual budget across competing needs like roads, sanitation, healthcare, and education based on impact scores. Use OOP to model budget categories, projects, and constraints. This tool can help city councils visualize trade-offs and make more transparent, data-informed budget decisions.

54. Pothole Detection & Repair Prioritization System Build a pothole reporting and prioritization system using classification on citizen-submitted reports (with severity/location) and graph-based routing to generate the most efficient repair crew dispatch routes. Use OOP to model road segments, reports, and repair crews with status tracking. This ensures limited road maintenance crews are deployed where they will have the greatest safety impact first.

55. Disaster Shelter Capacity & Allocation Optimizer Create a shelter allocation system using bipartite matching algorithms to assign displaced families to the nearest available shelters with sufficient capacity during floods or other disasters. Use OOP to model shelters, families, and resource needs (food, medical) as connected entities. This ensures fair, efficient, and rapid shelter allocation during crisis situations when manual coordination breaks down.

56. Adaptive Learning Path Recommender Build a learning path recommendation system using decision trees trained on student performance data (quiz scores, time spent, topic mastery) to suggest personalized next topics or remedial content. Use OOP to model students, topics, and learning modules as interconnected objects supporting different subjects. This addresses the one-size-fits-all limitation of traditional classroom teaching, especially valuable for government schools with high student-teacher ratios.

57. Dropout Risk Prediction System Design a dropout risk prediction tool using classification models trained on attendance, academic performance, and socio-economic indicator datasets to flag at-risk students early. Generate an explainable risk report to guide teacher and counselor intervention. Use OOP to model student records and risk-factor categories for extensibility. This supports government initiatives aimed at reducing school dropout rates, particularly among girls and economically disadvantaged students.

58. Scholarship Eligibility & Disbursement System Build a scholarship management system using an OOP-based rule validation engine that checks student eligibility against multiple criteria (income, caste category, academic performance, attendance) drawn from various government scheme rules. Use hashing for fast duplicate-application detection across large applicant pools. This reduces fraudulent claims and manual processing delays common in scholarship disbursement schemes.

59. Timetable Generation System Create an automated timetable generation system using graph coloring algorithms to solve the constraint satisfaction problem of scheduling classes, teachers, and rooms without conflicts. Allow configurable constraints (teacher availability, room capacity, subject frequency) modeled through OOP classes. This solves a genuinely painful administrative task colleges face every semester, saving significant manual effort.

60. Student Plagiarism Detection Tool Build a plagiarism detection tool using string-matching algorithms (KMP or Rabin-Karp) to compare submitted assignments against a reference corpus and flag suspiciously similar sections. Compute a similarity score and highlight matched text spans for instructor review. Use OOP to model documents, comparison reports, and similarity thresholds. This helps maintain academic integrity in an era of easy content copying.

61. Skill-Gap Analyzer Design a skill-gap analysis tool that clusters student skill profiles (from resumes/self-assessments) and compares them against current job market demand data scraped from public job portals, using classification to identify missing in-demand skills. Use OOP to model students, skills, and job requirements. This helps career counseling cells guide students toward targeted upskilling before placements.

62. Rural Digital Literacy Tracker Build a digital literacy assessment and tracking system using classification models on survey data (device usage, internet access, digital task competency) to measure progress across rural communities enrolled in literacy programs. Use OOP to model individuals, assessment modules, and progress records. This helps government digital literacy missions measure real outcomes rather than just enrollment numbers.

63. Exam Seating Arrangement Optimizer Create an exam seating arrangement system using graph coloring/backtracking algorithms to ensure no two students from the same class or with similar roll numbers sit adjacent, minimizing malpractice opportunities across multiple exam halls. Use OOP to model halls, seats, and students with constraint rules. This automates a currently manual and error-prone administrative task for large-scale examinations.

64. Library Book Recommendation System Build a book recommendation engine using collaborative filtering basics (user-based or item-based similarity) on library borrowing history datasets to suggest relevant books to students. Use OOP to model books, students, and borrowing records. This can increase library engagement and help students discover relevant academic and extracurricular reading material more effectively than static catalog browsing.

65. Peer Tutoring Matchmaking System Design a peer tutoring matchmaking system using graph-based compatibility scoring that pairs students needing help in specific subjects with peer tutors strong in those subjects, factoring in schedule availability. Use OOP to model students, subjects, and matching sessions. This creates a low-cost, scalable academic support system, especially useful in resource-constrained institutions lacking sufficient faculty for remedial classes.

66. Placement Prediction System Build a placement outcome prediction model using logistic regression trained on academic performance, skill certifications, and mock interview scores to estimate a student's placement readiness probability. Provide explainable feedback on improvement areas. Use OOP to model student profiles and placement criteria. This helps training and placement cells prioritize intervention for students most at risk of remaining unplaced.

67. E-Learning Content Difficulty Classifier Create a text complexity classifier using basic NLP feature extraction (sentence length, vocabulary complexity, readability scores) to automatically tag e-learning content by difficulty level, helping match content to appropriate student levels. Use OOP to model content modules and difficulty metadata. This supports adaptive e-learning platforms in organizing large content libraries more usefully for diverse learners.

68. Attendance Anomaly Detector Build an anomaly detection system that analyzes biometric/log-based attendance patterns using statistical outlier detection to flag suspicious patterns like proxy attendance or unusual check-in/check-out sequences. Use OOP to model attendance records, students, and anomaly flags. This helps institutions maintain attendance integrity, particularly relevant for compliance with regulatory attendance requirements.

69. Career Path Recommendation Engine Design a career recommendation system using decision tree or random forest models trained on student interests, aptitude scores, and academic strengths to suggest suitable career paths and higher education options. Use OOP to model career categories, required skills, and student profiles. This provides much-needed structured career guidance often unavailable in government and rural schools.

70. Government School Resource Allocation Optimizer Build a resource allocation tool using basic linear programming to optimally distribute limited government education budgets across schools based on need indicators like student count, infrastructure deficiency, and teacher shortage data. Use OOP to model schools, resources, and allocation rules. This supports education departments in making more equitable, data-driven budget distribution decisions across districts.

71. Deforestation Rate Tracker Build a deforestation trend tracker using time-series regression on publicly available forest cover datasets to estimate rate of loss by region and project future cover under current trends. Visualize year-over-year change to support conservation policy advocacy. Use OOP to structure regions, datasets, and analysis pipelines. This provides evidence-based insight for forest departments and environmental NGOs working on conservation planning.

72. Carbon Footprint Calculator App Design a personal/organizational carbon footprint calculator using an OOP-based activity classification system (transport, electricity, diet, waste) where each activity type computes emissions using standard conversion factors. Aggregate and visualize footprint trends over time to encourage behavior change. This promotes environmental awareness and can be extended for corporate sustainability reporting.

73. Renewable Energy Output Predictor Build a solar/wind energy output prediction model using regression analysis on historical weather data (irradiance, wind speed, temperature) to forecast expected power generation for a given installation. Use OOP to model different renewable energy source types with distinct prediction logic. This helps energy planners and grid operators better balance renewable supply with demand, improving grid stability.

74. Wildlife Migration Pattern Tracker Create a wildlife movement modeling tool using graph-based pathway analysis on (simulated or public) GPS tracking datasets to identify common migration corridors and seasonal movement patterns of a species. Use OOP to model animals, tracking points, and habitat zones. This supports wildlife conservation authorities in identifying and protecting critical migration corridors from human encroachment.

75. River Water Quality Classifier Build a water quality classification system using machine learning models trained on chemical parameters (BOD, DO, pH, coliform count) from public river monitoring datasets to categorize water quality into usability grades. Use OOP to model monitoring stations, samples, and classification results. This supports pollution control boards in prioritizing river stretches needing urgent remediation.

76. Plastic Waste Segregation Efficiency Tracker Design a tracking system that uses classification algorithms on waste composition audit data to measure segregation efficiency (wet/dry/recyclable) across different city wards, identifying areas needing awareness campaigns. Use OOP to model wards, waste categories, and audit records. This supports municipal solid waste management compliance with segregation-at-source regulations.

77. Urban Tree Canopy Mapping & Green Cover Analyzer Build a green cover analysis tool using clustering algorithms on GIS/satellite-derived vegetation index data to map tree canopy density across city zones and identify areas needing afforestation. Use OOP to model zones, canopy metrics, and planting recommendations. This supports urban forestry departments in data-driven green cover expansion planning to combat urban heat islands.

78. Electric Vehicle Charging Station Placement Optimizer Create a facility-location optimization tool using algorithms to determine optimal EV charging station placement based on traffic density, existing station coverage gaps, and population data, maximizing accessibility while minimizing redundant coverage. Use OOP to model stations, coverage zones, and demand points. This supports the growing need for EV infrastructure planning in expanding Indian cities.

79. Landfill Capacity Forecasting System Build a landfill capacity forecasting tool using time-series analysis on waste generation trends and current fill-rate data to predict when a landfill will reach capacity, supporting proactive planning for new sites or waste-reduction campaigns. Use OOP to model landfills, waste inflow records, and capacity thresholds. This helps municipal bodies avoid the crisis-mode landfill overflow situations common in many Indian cities.

80. Rainwater Harvesting Potential Estimator Design a rainwater harvesting potential calculator using regression on rooftop area and regional rainfall datasets to estimate annual harvestable water volume for individual buildings or entire neighborhoods. Use OOP to model buildings, catchment areas, and harvesting systems. This tool can support municipal rainwater harvesting mandate compliance and promote household-level water conservation.

81. Coastal Erosion Prediction System Build a coastal erosion trend predictor using regression analysis on historical shoreline position datasets to project future erosion rates for vulnerable coastal stretches. Visualize risk zones to support coastal zone management planning. Use OOP to structure coastal segments, historical measurements, and risk projections. This is highly relevant for India's extensive and increasingly vulnerable coastline communities.

82. Energy Consumption Anomaly Detector Create a household energy anomaly detection system using clustering and outlier detection on simulated smart-meter consumption data to flag unusual usage spikes that may indicate faulty appliances, theft, or wastage. Use OOP to model households, meters, and consumption records.

This supports utility companies and consumers in identifying inefficiencies and reducing unnecessary electricity costs.

83. Biodiversity Index Calculator Build a biodiversity assessment tool using statistical aggregation methods on species-count survey datasets from a given region to compute standard biodiversity indices (like Shannon diversity index) and track ecological health over time. Use OOP to model species, survey records, and regional biodiversity profiles. This supports ecological researchers and forest departments in monitoring conservation program effectiveness.

84. Air Pollution Source Attribution System Design a pollution source attribution tool using correlation and regression analysis to statistically link AQI spikes with likely contributing sources (vehicular traffic, industrial activity, stubble burning) based on time-of-day and seasonal patterns in public datasets. Use OOP to structure pollution sources and attribution models. This provides actionable evidence for targeted pollution control policy interventions.

85. Recycling Center Route Optimizer Build a route optimization tool using shortest-path algorithms to plan efficient collection routes for recyclable materials from multiple drop-off points to a central recycling facility, minimizing fuel and time costs. Use OOP to model collection points, vehicles, and recyclable material categories. This improves the economic viability and efficiency of community recycling programs.

86. Forest Fire Spread Predictor Create a forest fire spread simulation using a cellular automaton model combined with graph-based propagation logic, factoring in wind direction, vegetation density, and terrain slope from public datasets to predict fire spread patterns. Use OOP to model terrain cells and fire-state transitions. This supports forest departments in fire response planning and resource pre-positioning during high-risk seasons.

87. Sustainable Fishing Quota Allocator Design a fishing quota allocation tool using constraint-based optimization to distribute sustainable catch limits among fishing communities based on fish population data and historical catch records, preventing overfishing while protecting livelihoods. Use OOP to model fishing zones, communities, and quota records. This addresses the delicate balance between marine conservation and coastal community economic needs.

88. Microplastic Contamination Mapping Build a microplastic contamination mapping tool using spatial clustering on water/soil sample datasets to identify contamination hotspots along rivers or coastlines. Visualize concentration patterns to guide targeted cleanup and source-reduction efforts. Use OOP to model sample sites, contamination levels, and monitoring records. This raises data-driven awareness of an increasingly critical but under-monitored pollution issue.

89. Green Building Certification Scorer Create a green building scoring tool using an OOP-based rule/criteria engine that evaluates buildings against sustainability parameters (energy efficiency, water conservation, materials, waste management) to generate a certification-readiness score. Use weighted scoring logic similar to real certification standards like GRIHA/IGBC. This helps architects and builders self-assess sustainability compliance before formal certification processes.

90. Climate-Resilient Crop Advisory System Build a climate-resilient farming advisory system combining regression-based climate trend analysis with rule-based crop-suitability logic to recommend climate-adapted crop varieties and practices for a given region's projected future climate. Use OOP to model crops, climate zones, and advisory rules. This helps farmers proactively adapt to shifting weather patterns rather than reactively responding to crop failures.

91. Microfinance Loan Default Risk Predictor Build a loan default risk prediction model using classification algorithms trained on microfinance borrower datasets (repayment history, income proxy indicators, loan purpose) to help microfinance institutions make more informed lending decisions. Use OOP to model borrowers, loans, and risk profiles. This supports financial inclusion by helping lenders extend credit responsibly to underserved populations without excessive collateral requirements.

92. Personal Budget Tracker & Expense Classifier Design a personal finance tracker that uses OOP-based transaction modeling combined with basic NLP techniques to automatically categorize expenses from transaction descriptions (groceries, utilities, entertainment). Generate spending trend visualizations and savings recommendations. This promotes financial literacy and better money management among students and first-time earners.

93. Rural Banking Transaction Fraud Detector Build a fraud detection system using anomaly detection algorithms on simulated banking transaction datasets to flag unusual patterns (sudden large withdrawals, atypical transaction locations/times) common in rural banking fraud cases. Use OOP to model accounts, transactions, and fraud alerts. This helps protect vulnerable, often less digitally-literate rural banking customers from financial fraud.

94. Digital Wallet Transaction Routing Optimizer Create a transaction routing optimization system using graph-based network algorithms to select the fastest, lowest-cost path for digital payment transactions across multiple banking/payment gateway nodes. Use OOP to model payment gateways, transaction routes, and cost/latency metrics. This addresses the reliability and speed challenges faced by digital payment systems during high-load periods like festival sales.

95. Credit Score Simulation Engine Build a credit score simulation tool using a weighted, rule-based scoring engine (implemented via OOP strategy pattern) that estimates creditworthiness based on income, repayment history, and existing debt, mirroring how real credit bureaus operate. Allow users to simulate how financial decisions would affect their score. This promotes financial literacy about credit systems, especially for first-time credit seekers.

96. Stock Market Trend Predictor Design a stock trend analysis tool using time-series regression and moving-average techniques on historical stock price datasets to identify trend patterns and generate simple buy/hold/sell signals for educational purposes. Use OOP to model stocks, price histories, and trading strategies. This helps students understand basic quantitative finance concepts through hands-on data analysis rather than pure theory.

97. Self-Help Group (SHG) Savings & Loan Tracking System Build a digital ledger system for SHGs using an OOP inheritance model to track member savings, internal loans, and repayments transparently, replacing error-prone manual registers commonly used in rural SHGs. Generate automated statements and default alerts. This directly supports women-led SHG financial management, a critical rural livelihood mechanism across India.

98. Insurance Premium Calculator Create an insurance premium calculation engine using an actuarial rule-based decision system (OOP design) that computes premiums based on risk factors like age, health indicators, and coverage type, similar to real insurance underwriting logic. This helps demystify insurance pricing for consumers and can serve as a foundation for comparing policy options transparently.

99. Tax Filing Anomaly/Fraud Detector Build an anomaly detection tool that flags inconsistent or suspicious patterns in simulated tax filing datasets (mismatched income vs expenditure, duplicate deductions) using statistical and classification-based methods. Use OOP to model filings, taxpayers, and flagged anomalies. This demonstrates how tax authorities can use data science to improve compliance monitoring and reduce revenue leakage.

100. Investment Portfolio Diversification Optimizer Design a portfolio optimization tool using dynamic programming to allocate a fixed investment budget across asset categories (equity, debt, gold) to maximize expected returns within a given risk tolerance, based on historical return and volatility datasets. Use OOP to model asset classes and portfolio strategies. This introduces students to practical quantitative investment concepts through a working simulation.

101. Cryptocurrency Price Volatility Analyzer Build a volatility analysis tool using statistical time-series methods (standard deviation, rolling volatility) on public cryptocurrency price datasets to help users understand risk levels before investing. Visualize volatility trends alongside price movements. Use OOP to model different cryptocurrencies and analysis metrics. This promotes informed, risk-aware financial decision-making among young, increasingly crypto-curious investors.

102. ATM Cash Replenishment Optimizer Create an ATM cash management system using demand forecasting on historical withdrawal patterns combined with priority scheduling to optimize replenishment routes and timing, minimizing both cash-out incidents and idle cash. Use OOP to model ATMs, cash-in-transit vehicles, and replenishment schedules. This reduces operational costs for banks while improving customer service reliability.

103. Gig-Economy Worker Payment Reconciliation System Build a payment reconciliation system for gig-economy platforms using an OOP-based transaction ledger that tracks worker earnings across multiple jobs, platform fees, and payout schedules transparently. Flag discrepancies using automated reconciliation logic. This addresses growing concerns about payment transparency and fair compensation for India's expanding gig workforce.

104. Financial Literacy Quiz Adaptive Engine Design an adaptive financial literacy quiz system using decision-tree-based difficulty adjustment that personalizes question difficulty based on a learner's ongoing performance, similar to adaptive testing platforms. Use OOP to model quiz modules, questions, and learner progress. This supports scalable financial literacy education for populations with varying baseline knowledge levels.

105. Peer-to-Peer Lending Matchmaking System Build a P2P lending platform backend using graph-based risk-compatibility scoring to match lenders with borrowers based on risk appetite, loan amount, and repayment terms. Use OOP to model lenders, borrowers, loans, and matching logic. This creates an alternative credit access channel for borrowers underserved by traditional banking, while giving lenders more informed matching options.

106. Flood Evacuation Route Planner Build an evacuation route planning tool using shortest-path algorithms on a real or simulated terrain/road-network graph to compute the fastest safe evacuation routes from flood-prone zones to designated shelters, accounting for blocked or submerged roads. Use OOP to model roads, shelters, and hazard zones with dynamic states. This supports disaster management authorities in generating real-time evacuation guidance during flood emergencies.

107. Earthquake Damage Severity Predictor Design a damage severity prediction model using classification algorithms trained on seismic magnitude, building type, and soil condition data (from public earthquake datasets) to estimate likely damage levels across a region. Use OOP to model buildings, zones, and damage assessments. This helps disaster response teams prioritize rescue and resource deployment to the most severely affected areas first.

108. Emergency Resource Dispatch Optimizer Build an emergency resource dispatch system using priority queues combined with graph-based routing to allocate limited resources (rescue teams, medical supplies, food) to disaster-affected areas based on urgency and accessibility. Use OOP to model resources, requests, and dispatch units. This maximizes the impact of limited disaster response resources during the critical early hours of a crisis.

109. Landslide Risk Zone Classifier Create a landslide risk classification tool using regression analysis on rainfall intensity and slope-gradient datasets to identify high-risk zones along hilly roads and settlements. Generate early-warning alerts during heavy rainfall periods. Use OOP to model terrain segments and risk thresholds. This is particularly relevant for hilly regions of Andhra Pradesh and neighboring states prone to monsoon landslides.

110. Cyclone Path Prediction Visualizer Build a cyclone trajectory visualization tool using regression analysis on historical cyclone path datasets to project likely future paths and affected coastal regions based on current position and speed data. Use OOP to model cyclone tracking data and affected-zone predictions. This supports coastal disaster management authorities in issuing timely, region-specific evacuation advisories.

111. Disaster Relief Supply Chain Optimizer Design a relief supply distribution system using network flow algorithms to optimize the movement of essential supplies (food, water, medicine) from central depots to multiple disaster-affected distribution points with capacity and route constraints. Use OOP to model depots, distribution points, and supply inventories. This ensures faster, more equitable relief distribution during large-scale disasters.

112. Missing Person Search-and-Track System Build a missing person tracking system using graph-based last-seen network analysis to model reported sighting locations as connected nodes, helping investigators identify likely search zones based on movement patterns. Use OOP to model persons, sightings, and search zones. This supports police and disaster response teams in systematically organizing search efforts rather than relying on ad-hoc methods.

113. Building Structural Safety Scorer Create a structural safety assessment tool using a rule-based OOP inspection checklist system that scores buildings against safety parameters (age, material, maintenance history, code compliance) to flag high-risk structures needing urgent inspection. This supports municipal building safety departments in prioritizing limited inspection resources toward the most dangerous structures first.

114. Wildfire Evacuation Shelter Capacity Balancer Build a shelter capacity balancing tool using bipartite matching algorithms to distribute evacuees across available shelters based on capacity, distance, and special needs (medical, elderly, families), preventing overcrowding at any single location. Use OOP to model shelters and evacuee groups. This ensures more organized and humane shelter management during large-scale evacuation events.

115. Emergency Helpline Call Triage System Design a call triage system for emergency helplines using priority queue classification that ranks incoming calls by estimated urgency based on keyword and pattern analysis of call transcripts, ensuring critical calls are attended to first. Use OOP to model calls, operators, and triage categories. This improves response efficiency for helplines that frequently face call volume surges during emergencies.

116. Drought Early-Warning System Build a drought prediction system using time-series regression on rainfall deficit and reservoir level datasets to generate early warnings for regions likely to face water scarcity in upcoming seasons. Use OOP to model regions, water sources, and warning thresholds. This gives administrations crucial lead time to plan water rationing, crop advisories, and relief measures before a drought becomes a crisis.

117. Search-and-Rescue Drone Path Optimizer Create a drone path planning tool using graph traversal algorithms to compute optimal search patterns over a disaster-affected area, maximizing area coverage within battery/time constraints. Use OOP to model search grids, drones, and detected points-of-interest. This supports modern disaster response teams increasingly relying on drones for faster search-and-rescue operations in inaccessible terrain.

118. Community Disaster-Preparedness Scoring System Build a community preparedness assessment tool using a weighted rule engine (OOP design) that scores neighborhoods or villages on disaster-readiness parameters (evacuation plan awareness, emergency kit availability, shelter proximity). Use the scores to target awareness campaigns where preparedness is lowest. This shifts disaster management from purely reactive to proactive community-level resilience building.

119. Post-Disaster Infrastructure Damage Assessment Classifier Design an infrastructure damage classification tool using classification algorithms on structured damage-report data (or basic image features) to categorize damage severity across roads, bridges, and buildings after a disaster. Use OOP to model infrastructure assets and damage records. This speeds up damage assessment surveys that traditionally take weeks of manual inspection, accelerating relief fund allocation.

120. Multi-Hazard Risk Index Calculator Build a composite risk index calculator that combines multiple regression models (flood, earthquake, cyclone risk) into a single weighted multi-hazard vulnerability score for a given region, using publicly available hazard datasets. Use OOP to model hazard types and regional risk profiles. This supports integrated disaster risk reduction planning rather than siloed, single-hazard approaches.

121. Public Grievance Categorization & Routing System Build a grievance management system using NLP-based text classification to automatically categorize citizen complaints (water, roads, sanitation, corruption) and route them to the correct department using a rule-based OOP dispatcher. Track resolution time to generate departmental accountability reports. This improves transparency and efficiency in government grievance redressal, addressing widespread citizen frustration with slow bureaucratic responses.

122. Government Scheme Eligibility Checker Design an eligibility checking tool using an OOP-based rule engine that evaluates a citizen's profile against multiple government welfare scheme criteria simultaneously (income, category, age, occupation) to recommend all schemes they qualify for. This addresses the widespread problem of eligible citizens missing out on benefits simply due to lack of awareness about complex, fragmented scheme criteria.

123. Ration Distribution Fraud Detector Build an anomaly detection system that analyzes beneficiary transaction patterns at ration shops using statistical methods to flag suspicious activity like duplicate claims, ghost beneficiaries, or unusual quantity patterns. Use OOP to model beneficiaries, ration shops, and transaction records. This helps plug leakages in the Public Distribution System that divert subsidized food from genuine beneficiaries.

124. Voter Turnout Prediction System Create a voter turnout prediction tool using regression analysis on demographic and historical turnout datasets to forecast expected participation rates across constituencies, helping election commissions plan polling infrastructure and awareness campaigns for low-turnout areas. Use OOP to model constituencies, historical data, and predictions. This supports more efficient allocation of voter awareness resources.

125. Unemployment Trend Analyzer Build an unemployment trend analysis dashboard using time-series analysis on regional job statistics datasets to identify sectors and districts with rising unemployment, supporting targeted skill-development and job-creation policy planning. Use OOP to structure data sources and regional analysis modules. This gives policymakers timely, granular insight beyond aggregate national unemployment figures.

126. Elderly Pension Disbursement Tracker Design a pension tracking system using OOP-based transaction and state management to monitor disbursement status, flag delayed payments, and generate automated reminders for pending verifications. This addresses common pension delivery issues faced by elderly beneficiaries in rural areas, who often lack the means to follow up on bureaucratic delays themselves.

127. RTI (Right to Information) Request Routing & Tracking System Build an RTI management system using priority queues to track pending requests against statutory response deadlines, automatically flagging overdue requests for escalation. Use OOP to model requests, departments, and compliance status. This improves accountability in RTI processing, which frequently suffers from delays despite legal response-time mandates.

128. Waste-to-Resource Matching System Create a matching platform connecting NGOs, waste generators, and recyclers using graph-based matching algorithms that pair surplus resources (food, materials) with organizations that can use them, based on location and need type. Use OOP to model donors, recipients, and resource categories. This reduces wastage while supporting NGOs working on food security and resource redistribution.

129. Domestic Violence Helpline Call Pattern Analyzer Build an aggregate-level, fully anonymized call pattern analyzer for domestic violence helplines using clustering to identify temporal and regional trends (without exposing any individual case data), helping policymakers allocate support resources like shelters and counselors more effectively to high-need areas. Use OOP to structure aggregated statistical modules only, with strict privacy-by-design principles.

130. Public Toilet/Sanitation Facility Mapping & Quality Scorer Design a sanitation facility mapping tool using classification algorithms on facility condition survey data (cleanliness, water availability, functionality) to generate quality scores and identify facilities needing urgent maintenance. Use OOP to model facilities, inspections, and quality metrics. This supports Swachh Bharat-aligned sanitation infrastructure monitoring at the city or district level.

131. Migrant Worker Registration & Welfare Benefit Tracker Build a migrant worker welfare tracking system using an OOP inheritance hierarchy to model different worker categories and their eligible benefits (healthcare, housing, education for children) across state boundaries. This addresses the well-documented gap in welfare access faced by inter-state migrant workers who fall through jurisdictional cracks in benefit delivery systems.

132. Corruption Risk-Flagging System for Public Tenders Create an anomaly detection tool that flags unusual patterns in public tender/bid datasets (single-bidder tenders, price clustering near estimates, repeat winner patterns) using statistical outlier detection to support transparency audits. Use OOP to model tenders, bidders, and risk flags. This supports anti-corruption oversight bodies in prioritizing tenders for deeper investigation.

133. Census Data Anomaly & Inconsistency Detector Build a data validation tool using statistical validation algorithms to detect inconsistencies and anomalies in census or survey datasets (impossible age values, duplicate entries, logical contradictions) before they propagate into policy-making datasets. Use OOP to model records, validation rules, and error reports. This improves the reliability of foundational government datasets used for planning.

134. Community Resource-Sharing Library System Design a community tool/book/equipment sharing library platform using graph algorithms to match resource requests with available lenders based on proximity and availability, encouraging sustainable community resource use. Use OOP to model resources, members, and lending transactions. This promotes a sharing economy model that can reduce costs for economically constrained communities.

135. Government Job Vacancy vs Applicant Skill Matcher Build a matching system using classification and clustering to compare government job vacancy requirements against a pool of applicant skill profiles, identifying best-fit candidates and highlighting systemic skill gaps in the applicant pool. Use OOP to model jobs, applicants, and matching scores. This can make public sector recruitment more efficient and evidence-based.

136. Anganwadi Resource Allocation Optimizer Create a resource allocation tool using constraint-based optimization to distribute limited nutrition and educational resources across Anganwadi centers based on child enrollment numbers and need-severity indicators. Use OOP to model centers, resources, and allocation rules. This supports more equitable early childhood care resource distribution in a system that currently varies widely by local administrative capacity.

137. Public Transport Accessibility Scorer for Differently-Abled Citizens Design an accessibility scoring tool using a rule-based OOP scoring engine that evaluates public transport stops and vehicles against accessibility parameters (ramps, tactile paths, audio announcements) to generate an accessibility index for city planners. This supports evidence-based advocacy and planning for genuinely inclusive public transport infrastructure.

138. Water Tanker Distribution Scheduler for Drought-Hit Areas Build a water tanker scheduling system using greedy and priority algorithms to allocate limited water tanker trips to the most water-stressed localities first, based on reported shortage severity and population data. Use OOP to model tankers, localities, and distribution schedules. This ensures fairer water distribution during acute drought or supply-disruption periods.

139. NGO Fund Utilization Tracker & Anomaly Detector Create a fund utilization tracking system using classification models to flag unusual spending patterns in NGO financial disclosures that deviate from stated program budgets, supporting donor transparency and accountability. Use OOP to model funds, expenditures, and programs. This helps donors and regulators identify NGOs needing closer financial scrutiny.

140. Digital Identity Duplicate-Detection System Build a duplicate identity detection system using string similarity algorithms (edit distance, phonetic matching) to identify likely duplicate or fraudulent entries in a beneficiary database, even when names are misspelled or formatted differently. Use OOP to model identity records and match-confidence scores. This helps prevent duplicate benefit claims across large government welfare databases.

141. Phishing URL Detection System Build a phishing URL classifier using classification models (logistic regression/decision trees) trained on URL structural features (length, special characters, domain age proxies) from public phishing datasets to flag suspicious links in real time. Use OOP to model URLs, features, and classification results. This provides a practical, lightweight defense layer against a widespread and constantly evolving cyber threat.

142. Password Strength Analyzer & Breach-Checker Design a password security tool using hashing algorithms to check user passwords against known breach datasets without ever storing or transmitting the plaintext password, alongside a strength-scoring engine that evaluates entropy and common-pattern usage. Use OOP to model password policies and scoring rules. This promotes better personal cybersecurity hygiene through practical, privacy-respecting tooling.

143. Network Intrusion Detection Simulator Build a network intrusion detection simulator using anomaly detection algorithms on simulated network traffic log datasets to flag unusual patterns indicative of potential intrusions (port scanning, unusual data volume). Use OOP to model network nodes, traffic logs, and alert rules. This gives students hands-on experience with core cybersecurity monitoring concepts using real data science techniques.

144. Fake News/Misinformation Classifier Create a misinformation detection tool using NLP-based text feature analysis (source credibility signals, linguistic patterns, sensationalism markers) trained on public fake-news datasets to classify news articles as likely reliable or unreliable. Use OOP to model articles, sources, and classification confidence scores. This addresses the growing societal challenge of misinformation spread, particularly relevant during elections and public health crises.

145. Data Anonymization Tool for Public Datasets Build a data anonymization tool implementing the k-anonymity algorithm to generalize/suppress identifying attributes in datasets before public release, ensuring individual records cannot be re-identified while preserving analytical usefulness. Use OOP to model datasets, quasi-identifiers, and anonymization rules. This supports responsible open-data publishing practices for government and research datasets containing personal information.

146. Social Media Bot Account Detector Design a bot account detection tool using classification on behavioral pattern features (posting frequency, account age, follower-following ratio, content repetition) from public social media datasets to flag likely automated accounts. Use OOP to model accounts, behavior metrics, and detection scores. This helps address the growing problem of coordinated inauthentic behavior and misinformation amplification on social platforms.

147. Secure File-Sharing System with Access Control Build a secure file-sharing system using an OOP-based role hierarchy (admin, editor, viewer) to enforce granular access control permissions, combined with basic encryption for stored files. Use OOP inheritance and interfaces to model different user roles and permission sets cleanly. This demonstrates practical secure-by-design principles applicable to any organization handling sensitive documents.

148. Digital Footprint Privacy Risk Scorer Create a privacy risk assessment tool using a rule-based weighted analysis engine that evaluates a user's digital footprint (public social media info, data-sharing app permissions) to generate a privacy risk score with actionable recommendations. Use OOP to model risk factors and scoring categories. This promotes digital literacy around personal data privacy, an increasingly critical life skill.

149. SIM-Swap/Identity Fraud Pattern Detector Build an anomaly detection tool that identifies suspicious transaction sequences (rapid SIM change followed by banking OTP requests, unusual login locations) using pattern-based anomaly detection on simulated telecom/banking datasets. Use OOP to model accounts, events, and fraud-risk flags. This addresses a rapidly growing and financially devastating category of digital fraud in India.

150. Encrypted Community Bulletin Board System Design a secure community bulletin board system using cryptographic hashing for message integrity verification combined with an OOP design modeling users, posts, and moderation roles. Implement basic encryption for sensitive community postings (like neighborhood watch alerts). This demonstrates how secure, trustworthy digital community infrastructure can be built even with foundational-level cryptography concepts.